Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If the lines $3x - 4y - 7 = 0$ and $2x - 3y - 5 = 0$ are two diameters of a circle of area 49π square units, the equation of the circle is

(A) $x^2 + y^2 - 2x + 2y - 47 = 0$

(B) $x^2 + y^2 + 2x - 2y - 47 = 0$

(C) $x^2 + y^2 + 2x - 2y - 62 = 0$

(D) $x^2 + y^2 - 2x + 2y - 62 = 0$

Q.2

If the lines $2x + 3y + 1 = 0$ and $3x - y - 4 = 0$ lie along diameters of a circle of circumference 10π, then the equation of the circle is

(A) $x^2 + y^2 - 2x + 2y - 23 = 0$

(B) $x^2 + y^2 + 2x - 2y - 23 = 0$

(C) $x^2 + y^2 + 2x + 2y - 23 = 0$

(D) $x^2 + y^2 - 2x - 2y - 23 = 0$

Q.3

The intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$ is AB. Equation of the circle on AB as a diameter is

(A) $x^2 + y^2 - x - y = 0$

(B) $x^2 + y^2 + x - y = 0$

(C) $x^2 + y^2 + x + y = 0$

(D) $x^2 + y^2 - x + y = 0$

Q.4

The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3a$ is

(A) $x^2 + y^2 = 9a^2$

(B) $x^2 + y^2 = 16a^2$

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(C) $x^2 + y^2 = 4a^2$

(D) $x^2 + y^2 = a^2$

Q.5

A line is drawn through the point $\mathrm{P}(3, 11)$ to cut the circle $\mathrm{x}^2 + \mathrm{y}^2 = 9$ at A and B. Then PA .PB is equal to

(A) 9

(B) 121

(C) 205

(D) 139

Q.6

A square is inscribed in the circle $x^2 + y^2 - 2x + 4y + 3 = 0$, whose sides are parallel to the coordinate axes then, one vertex of the square is

(A) $(1 + \sqrt{2}, -2)$

(B) $(1 - \sqrt{2}, -2)$

(C) $(1, -2 + \sqrt{2})$

(D) None of these

Q.7

The equation of the circle which touches both the axes and whose radius is a, is

(A) $x^2 + y^2 - 2ax - 2ay + a^2 = 0$

(B) $x^2 + y^2 + ax + ay - a^2 = 0$

(C) $x^2 + y^2 + 2ax + 2ay - a^2 = 0$

(D) $x^2 + y^2 - ax - ay + a^2 = 0$

Q.8

If the centre of the circle passing through the origin is $(3, 4)$, then the intercepts cut off by the circle on x -axis and y -axis respectively are

(A) 3 unit and 4 unit

(B) 6 unit and 4 unit

(C) 3 unit and 8 unit

(D) 6 unit and 8 unit

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Q.9

If two circles A, B of equal radii pass through the centres of each other, then what is the ratio of the length of the smaller arc to the circumference of the circle A cut off by the circle B ?

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) $\frac{2}{3}$

Q.10

Two points P and Q are taken on the line joining the points $\mathrm{A}(0,0)$ and $\mathrm{B}(3 \mathrm{a}, 0)$ such that $\mathrm{AP}=\mathrm{PQ}=\mathrm{QB}$. Circles are
drawn on AP, PQ and QB as diameters. The locus of the point S, the sum of the squares of the tangents from which to the three circles is equal to b^2, is

(A) $x^2+y^2-3ax+2a^2-b^2=0$

(B) $3\left(x^2+y^2\right)-9ax+8a^2-b^2=0$

(C) $x^2+y^2-5ax+6a^2-b^2=0$

(D) $x^2+y^2-ax-b^2=0$

Q.11

Under which one of the following conditions does the circle $x^2+y^2+2gx+2fy+c=0$ meet the x -axis in two points on opposite sides of the origin?

(A) $\mathrm{c}>0$

(B) $c<0$

(C) $\mathrm{c}=0$

(D) $c \leq 0$

Q.12

The parametric form of equation of the circle $x^2+y^2-6x+2y-28=0$ is

(A) $x=-3+\sqrt{38}\cos\theta, y=-1+\sqrt{38}\sin\theta$

(B) $x=\sqrt{28}\cos\theta, y=\sqrt{28}\sin\theta$

(C) $x=-3-\sqrt{38}\cos\theta, y=1+\sqrt{38}\sin\theta$

(D) $x=3+\sqrt{38}\cos\theta, y=-1+\sqrt{38}\sin\theta$

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Q.13

The point diametrically opposite to the point $\mathrm{P}(1,0)$ on the circle $\mathrm{x}^2+\mathrm{y}^2+2\mathrm{x}+4\mathrm{y}-3=0$ is

(A) $(-3,4)$

(B) $(-3,-4)$

(C) $(3,4)$

(D) $(3,-4)$

Q.14

The point $([\mathrm{P}+1],[\mathrm{P}])$(where $[\mathrm{x}]$ is the greatest integer less than or equal to x), lying inside the region bounded by the circle $x^2+y^2-2x-15=0$ and $x^2+y^2-2x-7=0$, then

(A) $\mathrm{P} \in [-1,0) \cup [0,1) \cup [1,2)$

(B) $\mathrm{P} \in [-1,2)-\{0,1\}$

(C) $\mathrm{P} \in (-1,2)$

(D) none of these

Q.15

Let S is a circle with centre $(0,\sqrt{2})$. Then

(A) There cannot be any rational point on S

(B) There can be infinitely many rational points on S

(C) There can be at most two rational points on S

(D) There are exactly two rational points on S

Answers & Solutions

Q.1 Answer:

$x^2+y^2-2x+2y-47=0$

Solution:

Give diameters are $3x-4y-7=0$ **and** $2x-3y-5=0$
Solving we get centre $(1,-1)$ **and Area** $=\pi r^2=49\pi \Rightarrow r=7 \Rightarrow (\mathrm{x}-1)^2+(\mathrm{y}+1)^2=49$ **is equation**
$\Rightarrow x^2+y^2-2x+2y-47=0$

Q.2 Answer:

$x^2+y^2-2x+2y-23=0$

Solution:

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Let centre of circle be (h, k) and radius $= \mathrm{r}$

Circumference of circle $= 2\pi \mathrm{r}$

$2\pi \mathrm{r} = 10\pi$

$\mathrm{r} = 5$ units

Also, point (h, k) lies on both lines

$2h + 3k + 1 = 0$ and $3h - k - 4 = 0$

Solving these two equations

$2\,\mathrm{h} + 3\mathrm{k} + 1 = 0$

$9\,\mathrm{h} - 3\mathrm{k} - 12 = 0$

$11h - 11 = 0$

$\mathrm{h} = 1$ and $\mathrm{k} = -1$

Equation of circle becomes

$(x-1)^2 + (y+1)^2 = 5^2$

$x^2 + y^2 - 2x + 2y - 23 = 0$

Q.3 Answer:

$x^2 + y^2 - x - y = 0$

Solution:

The line $y = x$ meets the circle $x^2 + y^2 - 2x = 0$ when $2x^2 - 2x = 0$

$\Rightarrow x = 0$ or 1.

Hence, the equation of the required circle is $x(x-1) + y(y-1) = 0$

Q.4 Answer:

$x^2 + y^2 = 4a^2$

Solution:

As the centre of the circum circle of an equilateral triangle is its centroid and the distance of the centroid from the vertex of the triangle is $2/3$ of its median through that vertex. So the distane of the centre from a vertex of the triangle is $(2/3) \times 3\mathrm{a} = 2\mathrm{a}$ and hence the equation of the circum circle is $x^2 + y^2 = 4a^2$.

Q.5 Answer:

121

Solution:

Equation of any line through $(3, 11)$ is $\frac{x-3}{\cos\theta} = \frac{y-11}{\sin\theta} = r$ (say)

Then the coordinates of a point on this line at a distance r from $(3, 11)$ are

$(3 + \mathrm{r}\cos\theta, 11 + \mathrm{r}\sin\theta)$

and if this lies on the given circle

$\mathrm{x}^2 + \mathrm{y}^2 = 9$ then $(3 + r\cos\theta)^2 + (11 + r\sin\theta)^2 = 9$

$\Rightarrow 9 + 121 + 2r(3\cos\theta + 11\sin\theta) + r^2 = 9$

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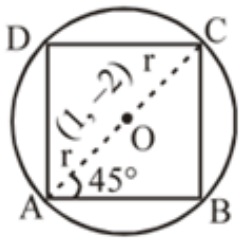
$\Rightarrow r^2 + 2r(3\cos\theta + 11\sin\theta) + 121 = 0$

which is quadratic in r, gives two values of r say r_1 and r_2 and hence the distances of the points A and B from P. Thus, $PA.PB = r_1 r_2 = 121$

Q.6 Answer:

None of these

Solution:



$r = \sqrt{1^2 + (-2)^2 - 3} = \sqrt{2}$

we know that $\frac{x-x_1}{\cos\alpha} = \frac{y-y_1}{\sin\alpha} = \pm r$ **corrdinates of points are**

$\frac{x-1}{1/\sqrt{2}} = \frac{y+2}{1/\sqrt{2}} = \pm\sqrt{2}$

$x = \pm 1 + 1$ **and** $y = \pm 1 - 2$

$x = 2 \quad y = -1$ **i.e.** $(2, -1)$

So, $x = 2 \quad y$

$x = 0 \quad y = -3 \quad (0, -3)$

Q.7 Answer:

$x^2 + y^2 - 2ax - 2ay + a^2 = 0$

Solution:

Given,

$\Rightarrow$ **circle of radius a touches both axis in 1st quadrant so its centre will be** (a, a).

$\therefore$ **Required equation**

$\Rightarrow (x-a)^2 + (y-a)^2 = a^2$

$\Rightarrow x^2 + a^2 - 2ax + y^2 + a^2 - 2ay = a^2$

$\Rightarrow x^2 + y^2 - 2ax - 2ay + 2a^2 - a^2 = 0$

$\Rightarrow x^2 + y^2 - 2ax - 2ay + a^2 = 0$

Q.8 Answer:

6 unit and 8 unit

Solution:

Equation of circle having radius r and centre $(3, 4)$ is $= (x-3)^2 + (y-4)^2 = r^2$

if it is passing through $(0, 0) \therefore \quad (0-3)^2 + (0-4)^2 = r^2$

$\Rightarrow r^2 = 25$

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putting $y = 0$ $x = 6$ **unit** = **interception** x **-axis intercept on** y **axis** (**putting** $x = 0$) **is** $y = 8$ **unit**

Q.9 Answer:

$\frac{1}{3}$

Solution:

When two circles A **and** B **of equal radii pass through the centres of each other, the angle made by arc of** B **at the centre of** B **is** 90°**. So, length of small arc of** $B = \frac{2\pi r 90^{\circ}}{360^{\circ}} = \frac{\pi r}{2}$

Hence, circumference of A cut off by the circle B

$= 2\pi r - \frac{\pi r}{2} = \frac{3\pi r}{2}$

$\therefore$ **Required ratio** $= \frac{\pi r/2}{3\pi r/2} = \frac{1}{3}$

Q.10 Answer:

$3\left(x^2 + y^2\right) - 9ax + 8a^2 - b^2 = 0$

Solution:

Points P **and** Q **are** $(a, 0)$ **and** $(2a, 0)$ **So, the circles are** $(x - 0)(x - a) + (y - 0)(y - 0) = 0 \Rightarrow x^2 + y^2 - ax = 0$

$(x - a)(x - 2a) + (y - 0)(y - 0) = 0 \Rightarrow x^2 + y^2 - 3ax + 2a^2 = 0$

$(x - 2a)(x - 3a) + (y - 0)(y - 0) = 0$

$\Rightarrow x^2 + y^2 - 5ax + 6a^2 = 0$

Now, S be (h, k) then $h^2 + k^2 - ah + h^2 + k^2 - 3ah + 2a^2 + h^2 + k^2 - 5ah + 6a^2 = b^2$

$\Rightarrow \quad 3\left(h^2 + k^2\right) - 9ah + 8a^2 = b^2$

Q.11 Answer:

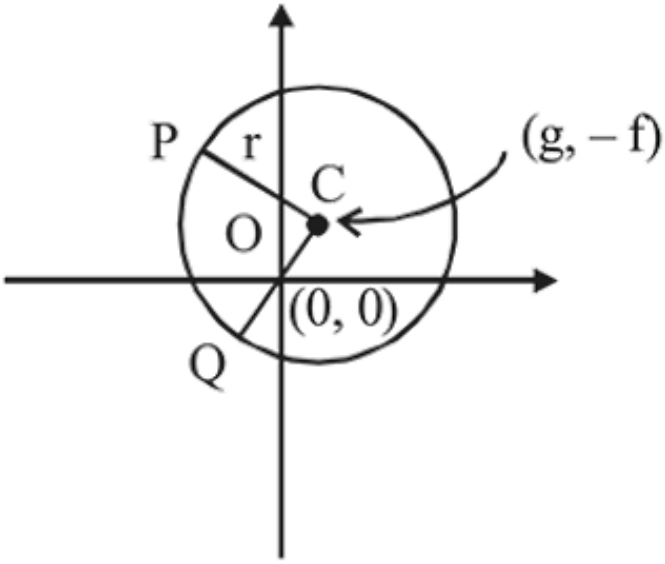
$c < 0$

Solution:

For a circle to meet x **-axis in two points on the opposite side of the origin its radius** r**, should be more the distance of its centre from the origin. Co-ordinate of centre of the circle** $x^2 + y^2 + 2gx$

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$+2fy + c = 0$ is $(-g, -f)$



In the figure shown, $OQ = OP = r$**, and distance of centre** C**, from origin,** O **is** $CO\ r > \sqrt{OC}$
i.e. $r > \sqrt{(-g)^2 + (-f)^2}$
or, $\sqrt{(-g)^2 + (-f)^2 - c} > \sqrt{(-g)^2 + (-f)^2}$
or, $g^2 + f^2 - c > g^2 + f^2$
or, $-c > 0$
or, $c < 0$

Q.12 Answer:

$x = 3 + \sqrt{38}\cos\theta, y = -1 + \sqrt{38}\sin\theta$

Solution:

Eq^{n} **of circle is**
$x^2 + y^2 - 6x + 2y - 28 = 0$
$2g = -6 \Rightarrow g = -3$ **and** $2f = 2 \Rightarrow f = 1$
$c = -28$
$\therefore r = \sqrt{g^2 + f^2 - c} = \sqrt{9 + 1 + 28}$
$= \sqrt{38}$
centre $:= (-g, -f) = (3, -1) = (h, k)$
$\therefore x = h + r \cdot \cos\theta$**, and** $y = k + r\sin\theta$
$\Rightarrow x = 3 + \sqrt{38}\cos\theta, y = -1 + \sqrt{38}\sin\theta$

Q.13 Answer:

$(-3, -4)$

Solution:

$x^2 + y^2 + 2x + 4y - 3 = 0$
Centre $\equiv (-1, -2)$

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Let point be $(h, k) \Rightarrow \frac{h+1}{2}=-1 \quad \frac{k+0}{2}=-2$
$\Rightarrow (\mathrm{h}, \mathrm{k}) \equiv (-3,-4)$

Q.14 Answer:

none of these

Solution:

Since the $([\mathrm{P}+1],[\mathrm{P}])$ **lies inside the circle** $x^2+y^2-2x-15=0[$ **But** $[x+n]=[x]+n, n \in N]$
$\therefore \quad [\mathrm{P}+1]^2+[\mathrm{P}]^2-2[\mathrm{P}+1]-15<0$
$([\mathrm{P}]+1)^2+[\mathrm{P}]^2-2([\mathrm{P}]+1)-15<0$
$2[\mathrm{P}]^2-16<0, \quad [\mathrm{P}]^2<8$
From the second circle $([\mathrm{P}]+1)^2+[\mathrm{P}]^2-2([\mathrm{P}]+1)-7>0$
$\Rightarrow \quad 2[\mathrm{P}]^2-8>0, \quad [\mathrm{P}]^2>4$
From $(1) \& (2), \quad 4<[\mathrm{P}]^2<8$, **which is not possible**
$\therefore$ **for no values of '** P **' the point will be within the region.**

Q.15 Answer:

There can be at most two rational points on S

Solution:

The equation of the circle S **is** $x^2+(y-\sqrt{2})^2=r^2$
Let the coordinates of any point on this circle be (h, k), **then**
$\mathrm{h}^2+(\mathrm{k}-\sqrt{2})^2=\mathrm{r}^2$
$\Rightarrow \mathrm{k}=\sqrt{2} \pm \sqrt{\mathrm{r}^2-\mathrm{h}^2}$
Since the above value of k **contains a constant irrational number** $\sqrt{2}$, **therefore, the only possible rational value of** k **is 0 . Hence,** $\sqrt{2} \pm \sqrt{r^2-h^2}=0 \Rightarrow r^2-h^2=2$
$\Rightarrow \mathrm{h}= \pm \sqrt{\mathrm{r}^2-2}$
Thus, we have following cases
(i) If r^2-2 **is a perfect square, there will be two rational points, viz.,** $\left(\sqrt{r^2-2}, 0\right)$ **and** $\left(-\sqrt{r^2-2}, 0\right)$ **on** S.
(ii) If r^2-2 **is not a perfect square, there will be no rational point on** S.
$\therefore$ **there can be at most two rational points on** S.

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